

Unknown sides



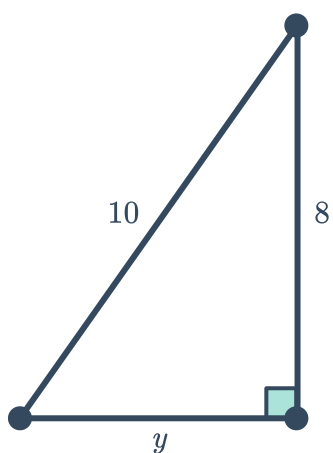
1 Find the number that completes a Pythagorean triple given the following:

- a The two largest numbers are 10 and 8.
- b The largest and smallest numbers are 41 and 9.

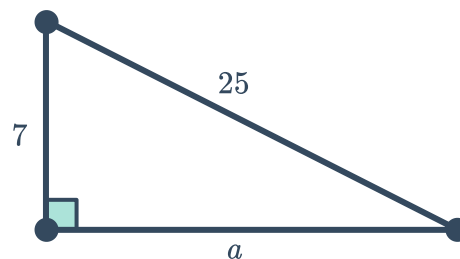
2 For the following right-angled triangles:

- i Write the equation that the sides of the triangle satisfy.
- ii Solve the equation to find the length of the unknown side.

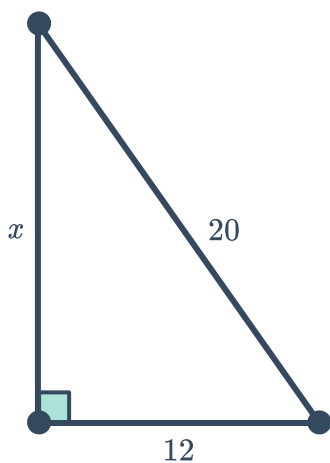
a



b

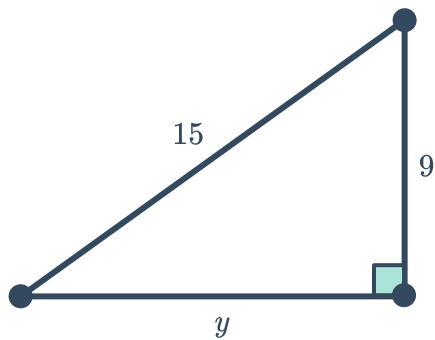


c

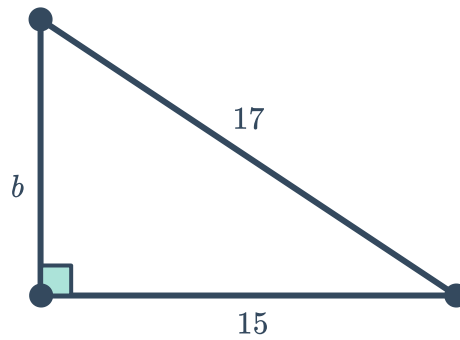


3 Find the length of the unknown side of the following triangles:

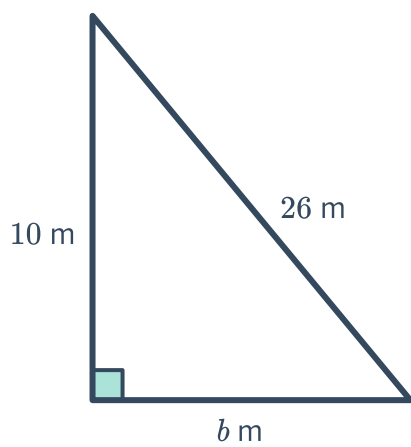
a



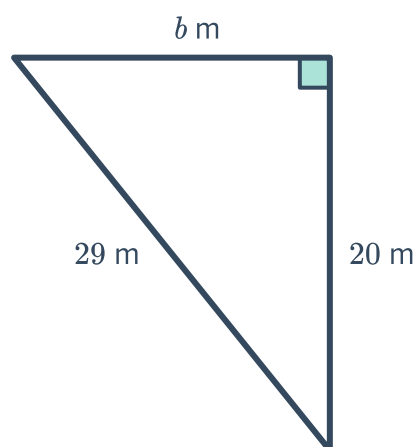
b



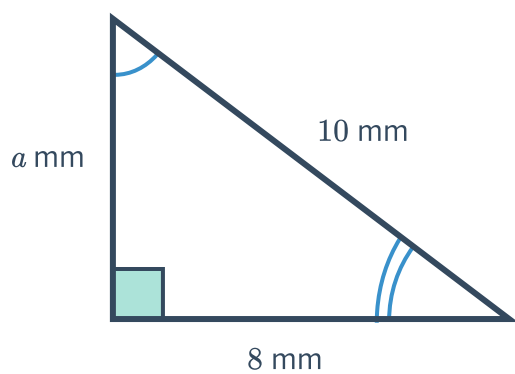
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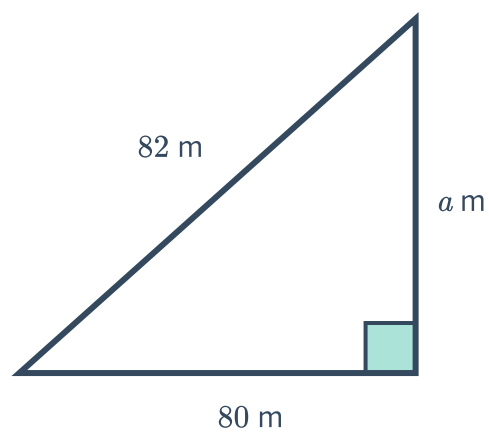
d



e

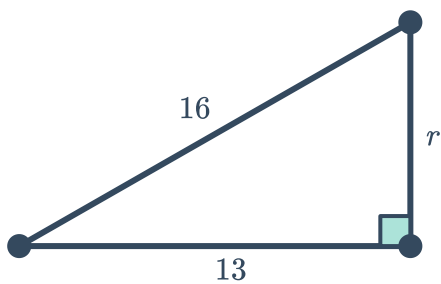


f

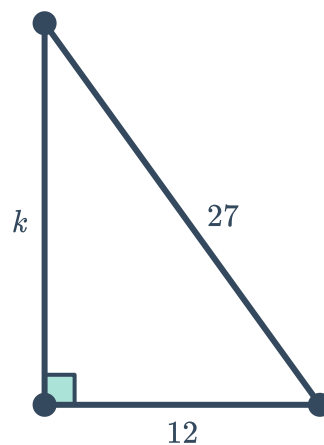


4 Find the length of the unknown side of the following triangles, correct to two decimal places:

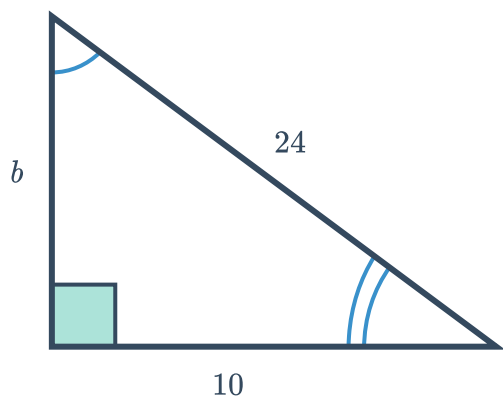
a



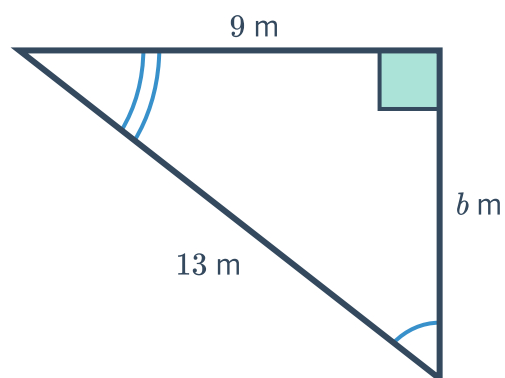
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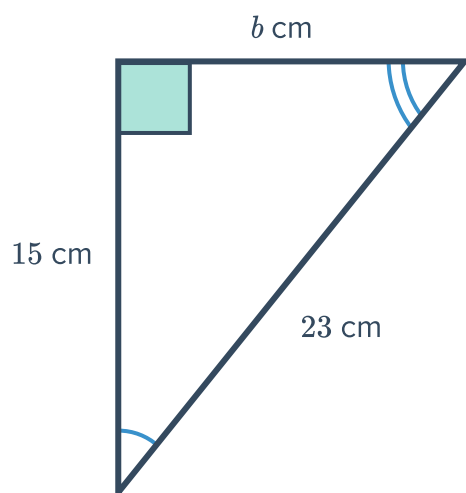
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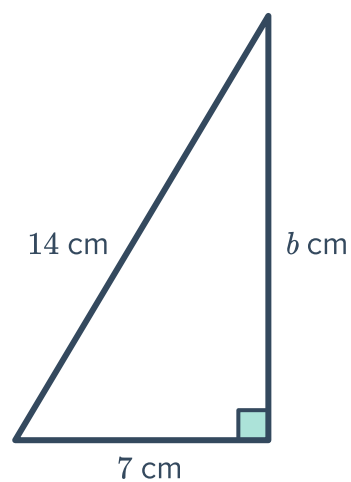
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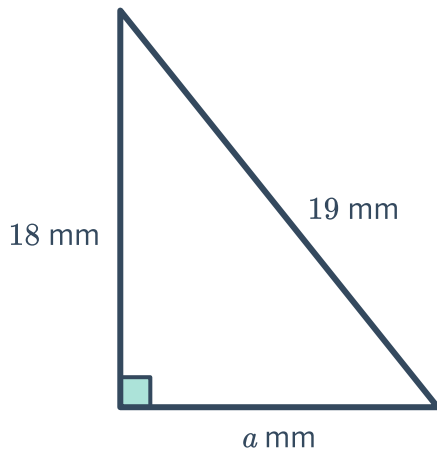
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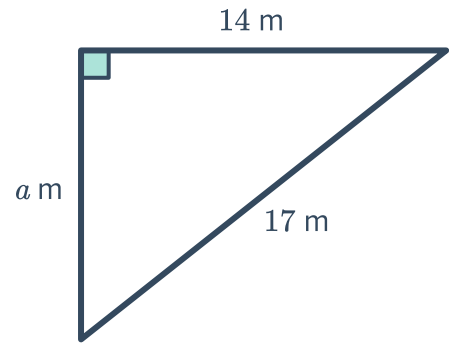
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g

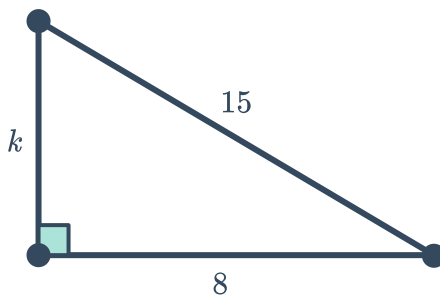


h

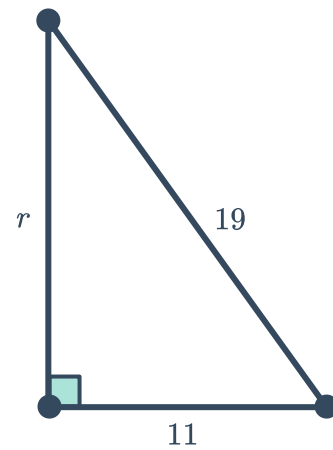


5 Find the length of the unknown side of the following triangles, giving your answer as a surd:

a

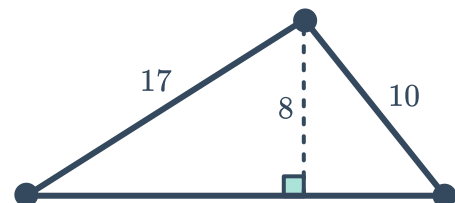


b



6 The two short sides of a triangle have lengths of 10 and 17. Sarah notices that she can divide the triangle into two smaller right-angled triangles, each with a height of 8.

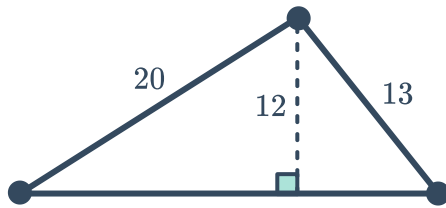
- Find the base lengths of the two right-angled triangles.
- Find the base length of the original triangle.



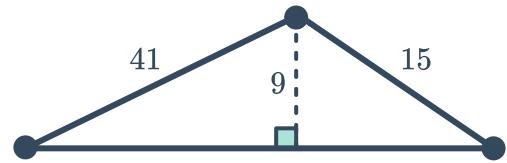
7 Find the base length of the following triangles.



a



b



Perimeter

8 Consider a right-angled triangle with hypotenuse of length 15 cm and another side of length 4 cm.

a Find the length of the other side, correct to two decimal places.

b Find the perimeter, correct to two decimal places.

9 Consider a rectangle with a diagonal of length 25 cm and a side of length 24 cm.

a Find the length of the other side.

b Find the perimeter.